EXHAUST SYSTEM - TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the exhaust system, refer to the relevant description and operation section of the workshop manual. REFER to: (309-00D Exhaust System - TDV8 4.4L Diesel)

Exhaust System (Description and Operation),
Diesel Particulate Filter (Description and Operation),
Diesel Particulate Filter (Description and Operation).

INSPECTION AND VERIFICATION

1. Verify the customer concern.

2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Diesel particulate filter ▪ Differential pressure sensor hoses ▪ Differential pressure sensor metal pipes and unions ▪ Differential pressure sensor mounting position integrity	▪ Differential pressure sensor ▪ Differential pressure sensor connector ▪ Differential pressure sensor wiring harness

▪ Metal fatigue ▪ Joints ▪ Mountings ▪ Clearance around components ▪ Damaged diesel particulate filter ▪ Gasket failure ▪ Flexible coupling ▪ Mass damper ▪ Exhaust valve ▪ Selective Catalytic Reduction (SCR) catalytic converter ▪ Flange connection – Low Pressure Exhaust Gas Recirculation (LPEGR) valve ▪ Flange connection - Primary turbocharger ▪ Flange connection - Secondary turbocharger	▪ Electrical connector(s) ▪ Sensor(s) ▪ Engine control module

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the symptom chart.

SYMPTOM CHART

DTC Index

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer approved diagnostic system)
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals

DTC	DESCRIPTION	POSSIBLE CAUSE	ACTION
P0030-29	HO2S Heater Control Circuit Bank 1 Sensor 1 - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a

		connector pin corrosion ▪ Heated oxygen sensor failure	new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0031-11	HO2S Heater Control Circuit Low Bank 1 Sensor 1 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Heated oxygen sensor heater circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0032-12	HO2S Heater Control Circuit High Bank 1 Sensor 1 - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Heated oxygen sensor heater circuit short circuit to power ▪ Connector is disconnected,	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required

		▪ connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0131-11	O2 Sensor Circuit Low Voltage Bank 1 Sensor 1 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Fuel injection related DTCs ▪ Air induction system leakage ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leakage ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0132-12	O2 Sensor Circuit High Voltage Bank 1 Sensor 1 - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leakage

		when another value was expected ▪ Fuel injection related DTCs ▪ Air induction system leakage ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0134-00	O2 Sensor Circuit No Activity Detected Bank 1 Sensor 1 - No sub type information	▪ Fuel injection related DTCs ▪ Air induction system leakage ▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Pre catalytic converter failure	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check air induction system for leakage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or

			corrosion ▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0135-29	O2 Sensor Heater Circuit Bank 1 Sensor 1 - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P014C-00	O2 Sensor Slow Response - Rich to Lean Bank 1 Sensor 1 - No sub type information	▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open

		disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P014D-00	O2 Sensor Slow Response - Lean to Rich Bank 1 Sensor 1 - No sub type information	▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0171-00	System Too Lean Bank 1 - No sub type information	▪ Other fuel injection related DTCs	▪ Check engine control module for fuel injection related DTCs and refer to

			relevant DTC index
P0172-00	System Too Rich Bank 1 - No sub type information	▪ Other fuel injection related DTCs	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index
P0234-77	Turbocharger/Supercharger "A" Overboost Condition - Commanded position not reachable	**NOTE:** A customer may express a concern that the amber Diesel Particulate Filter (DPF) warning indicator is illuminated on the Instrument Cluster (IC) requesting a DPF regeneration. If the regeneration is not completed the DPF warning indicator will then turn red. The DTCs **P2463-00 and P246B-00** will be stored in the Engine Control Module (ECM) memory ▪ The electric throttle air flow characteristics are incompatible with the engine calibration ▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its	▪ Check the engine control module for related DTCs and if DTCs **P2463-00 and P246B-00** are also logged GO to Pinpoint Test **A.** ▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Check exhaust system for mechanical integrity ▪ Check and correct the engine oil level ▪ Check air induction system / turbocharger for failure relating to excessive charged air conditions ▪ Check air induction system for leakage ▪ Check air induction system for excessive oil ▪ Check charge air pressure sensor for blockage ▪ Check turbocharger system / vanes for mechanical integrity ▪ Check electric throttle system / blade for mechanical integrity

			mechanical environment ▪ Excessive charged air pressure has been measured ▪ Other related DTCs ▪ Fuel injection related DTCs ▪ Exhaust system leakage ▪ Excessive engine oil level ▪ Excessive oil in the air induction system ▪ Air induction system leakage ▪ Charge air pressure sensor blocked ▪ Turbocharger system / vanes mechanical integrity ▪ Electric throttle system / blade mechanical integrity	▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
	P0420-00	Catalyst System Efficiency Below Threshold Bank 1 - No Sub Type Information		▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Check and install a new diesel oxidation catalyst (DOC). If the DPF is below the critical soot loading	

			NOTE:
			
			Diesel oxidation catalyst exothermic generation capability check. Diesel oxidation catalyst monitoring runs during the active regeneration of the particulate filter. The exhaust gas temperature measured at the catalyst outlet is compared against a model of the expected temperature increase during the active regeneration. A malfunction is reported when the measured temperature is significantly different to this model. The monitoring returns a judgment around 10min after the active particulate filter regeneration was started
			
			- Other related DTCs
			  - Airpath
			  - Manifold
			- Diesel oxidation catalyst failed, air path failures such as temperature and pressure sensors causing frequent and
				initiate a service regeneration. Confirm soot level reduces correctly. If the DPF is above the critical soot loading replace the DPF and reset the soot loading value
				
				- Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalyst circuit for short circuit to ground, short circuit to power, open circuit, high resistance
				- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
				- Check and install a new exhaust gas temperature sensor - Post catalyst as required
				- Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	

			hot regeneration, any air path failure which will lead to increased soot loading of the DPF such as charge air leak, or blocked air filter causing frequent and hot regeneration ▪ Exhaust gas temperature sensor - Post catalyst circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalyst failure	

	P0544-22	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 1 - Signal amplitude > maximum	**NOTE:**
			To monitor for too high temperature in the turbine housing; component protection. Temperature sensor temperature above threshold (899.96°C)
			▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high
			▪ Other related DTCs
				▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
				  ▪ 0x05CA Exhaust Turbine A Inlet Temperature
				  ▪ Check temperature sensor for plausibility with engine off
				  ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage
				  ▪ Check sensor for voltage plausibility
				▪ Check engine control module for

		▪ Airpath ▪ Turbocharger ▪ EGR ▪ Fuel system ▪ Turbocharger housing mechanical integrity ▪ Fuel injector (s) mechanical integrity ▪ Cylinders 1, 2 and 3 ▪ Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure	related DTCs and refer to relevant DTC index ▪ Check turbocharger housing for mechanical integrity ▪ Check fuel injector (s) for mechanical integrity ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Turbine temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0545-16	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out,	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas

		connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure	temperature sensor - Turbine temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0546-17	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Turbine temperature sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P064D-00	Internal Control Module O2 Sensor Processor Performance Bank 1 - No sub type information		▪ Check engine control module for charged air system related DTCs and refer to relevant DTC index ▪ Check engine control module for fuel system related

		NOTE: SPI communication within heated oxygen sensor internal chip has been disrupted - Other charged air system related DTCs - Other fuel system related DTCs - Air induction system leakage - EGR system leakage - Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure - Engine control module failure	DTCs and refer to relevant DTC index - Check air induction system for leakage - Check EGR system for leakage - Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new heated oxygen sensor as required - Check and install a new engine control module as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P2002-00	Diesel Particulate Filter Efficiency Below Threshold (Bank 1) - No sub type information	- Exhaust system leakage
		- Diesel particulate filter degraded
		- In cold climates differential pressure sensor hose lines or metal pipes may be frozen
		- Differential pressure sensor hoses connected incorrectly
			- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  - 0xD922 Particulate Filter Differential Pressure - Measured
			  - Measure with

			- Differential pressure sensor hoses crushed, blocked, split - Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Differential pressure sensor failure	engine off - Check differential pressure sensor hose lines are not frozen - Check differential pressure sensor hose lines are installed correctly - Check differential pressure sensor hose lines for mechanical integrity - Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new differential pressure sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
	P2031-22	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 2 - Signal amplitude > maximum		- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals - 0x03F7 Exhaust Gas Temperature	

			NOTE:
			To monitor for too high temperature at the diesel oxidation catalyst outlet location; DOC/DPF component protection. Temperature sensor temperature above threshold (874.96°C)
			▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high
			▪ Overfilled DPF during drop to idle with active DPF regeneration on
			▪ Blocked front face of DPF
			▪ Stuck closed electric throttle butterfly valve during active DPF regeneration
			▪ Other related DTCs
			  ▪ Airpath
			  ▪ Turbocharger
			  ▪ EGR
			  ▪ Fuel system
			▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance
				Bank2 Sensor 1
				  ▪ 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage
				  ▪ 0x042C Diesel Particulate Filter Soot Concentration
				▪ Check DPF for mechanical integrity
				▪ Check electric throttle butterfly valve for mechanical integrity
				▪ Check engine control module for related DTCs and refer to relevant DTC index
				▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
				▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required
				▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	

		- Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Pre catalytic converter failure	
P2032-16	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 2 - Circuit voltage below threshold	**NOTE:** Exhaust gas temperature sensor voltage below threshold (80mV) - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Pre catalytic converter failure	- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2033-	Exhaust Gas Temperature		Diagnosis of this

17	Sensor Circuit High Bank 1 Sensor 2 - Circuit voltage above threshold	**NOTE:** Exhaust gas temperature sensor voltage above threshold (4990mV) ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Pre catalytic converter failure	DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2080-62	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1 - Signal compare failure		▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CA Exhaust Turbine A Inlet Temperature ▪ Inspect connectors for signs of water ingress, and pins for damage and/or

			NOTE: Exhaust gas temperature sensor - Turbine temperature sensor rationality check. Temperature sensor is checked against other temperature sensors at coldstart - The engine control module detected failure when comparing two or more input parameters for plausibility - Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Turbine temperature sensor failure	corrosion - Check and install a new exhaust gas temperature sensor - Turbine temperature sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
	P2080-64	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1 - Signal plausibility failure		- Check engine control module for related DTCs and refer to relevant DTC index - Diagnosis of this	

			NOTE:
			
			Exhaust gas temperature sensor - Turbine temperature sensor rationality check during a driving cycle is not plausible (compared against a temperature model). Exhaust temperature sensor temperature is checked during a vehicle driving cycle against a model temperature for rationality. Some engine speed/load conditions apply. If the difference between the sensor temperature and the model exceeds a defined threshold, a malfunction is raised
			
			- The engine control module detected plausibility failures
			- Other related DTCs
			  - Airpath
			  - Manifold
			  - Fuel system
			- Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground,
				DTC may require using the manufacturer approved diagnostic system check datalogger signals
				  - 0x05CA Exhaust Turbine A Inlet Temperature
				- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
				- Check and install a new exhaust gas temperature sensor - Turbine temperature sensor as required
				- Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	

		short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure	

P2084-62	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 2 - Signal compare failure	**NOTE:**
		Exhaust gas temperature sensor temperature at least 50°C above other exhaust gas temperature sensors temperature after 90 minutes engine off time
		▪ The engine control module detected failure when comparing two or more input parameters for plausibility
		▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1
			  ▪ 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage
			▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required
			▪ Using the

		▪ Exhaust gas temperature sensor - Pre catalytic converter failure	manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P2084-64	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 2 - Signal plausibility failure	**NOTE:**
		
		Exhaust gas temperature sensor temperature is checked during a vehicle driving cycle against a model temperature for rationality. Some engine speed/load conditions apply. If the difference between the sensor temperature and the model exceeds a defined threshold, the DTC is set
		
		▪ The engine control module detected plausibility failures
		▪ Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Exhaust gas
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1
			  ▪ 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage
			▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required
			▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		temperature sensor - Pre catalytic converter failure	
P20EE-00	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - No sub type information	**NOTE:** Monitors SCR Nox catalyst efficiency - Reduced catalyst Nox efficiency ▪ SCR - Reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	**NOTE:** Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) ▪ Check for damaged or removed SCR catalyst ▪ Check for deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2195-00	O2 Sensor Signal Biased/Stuck Lean - Bank 1, Sensor 1 - No sub type information		▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index

			NOTE: Measured oxygen concentration implausibly high compared to model oxygen concentration for Sensor 1 Bank 1, during part load and during low part load ▪ Other related DTCs ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
	P2195-85	O2 Sensor Signal Biased/Stuck Lean - Bank 1, Sensor 1 - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Other related DTCs ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected,	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors

		connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2196-00	O2 Sensor Signal Biased/Stuck Rich - Bank 1, Sensor 1 - No sub type information	**NOTE:** Measured oxygen concentration implausibly low compared to model oxygen concentration for Sensor 1 Bank 1, during part load and during low part load ▪ Other related DTCs ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P2196-84	O2 Sensor Signal Biased/Stuck Rich - Bank 1, Sensor 1 - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Other related DTCs ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Check engine control module for fuel injection related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P223C-00	O2 Sensor Pumping Current Range/Performance Bank 1 - No sub type information	▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required

			▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2251-13	O2 Sensor Negative Current Control Circuit / Open - Bank 1, Sensor 1 - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P226D-00	Particulate Filter Deteriorated/Missing Substrate Bank 1 - No sub type information	▪ Other DPF related DTCs ▪ Diesel particulate substrate is missing - Severely damaged or removed ▪ DPF failure	

			NOTE: Operational requirements needed to allow the monitor to be fully tested. Drive at steady 70mph 6th gear for 10 minutes. Return to workshop and turn engine off, wait 5 minutes, key on and check for DTC ▪ Check engine control module for DPF related DTCs and refer to relevant DTC index ▪ Remove the DPF and visually inspect for severe damage and correct installation ▪ Check and install a new DPF as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2414-00	O2 Sensor Exhaust Sample Error Bank 1, Sensor 1 - No sub type information		▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		NOTE: Measured oxygen concentration implausible compared to other Nox Sensor, bank 1, during part load ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new heated oxygen sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P242A-22	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 3 - Signal amplitude > maximum	**NOTE:**
		To monitor for too high temperature at the diesel oxidation catalyst outlet location; DOC/DPF component protection. Temperature sensor temperature above threshold (874.96°C)
		▪ The engine control module measured a signal voltage above a
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2
			  ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage
			  ▪ 0x042C Diesel Particulate Filter Soot Concentration
			▪ Refer to the electrical circuit

		specified range but not necessarily a short circuit to power, gain too high
		▪ Exhaust gas temperature sensor - Post catalytic converter
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Exhaust gas temperature sensor - Post catalytic converter failure
		▪ Overfilled DPF during drop to idle with active DPF regeneration on
		▪ Blocked front face of DPF
		▪ Stuck closed electric throttle butterfly valve during active DPF regeneration
		▪ Other related DTCs
		  ▪ Airpath
		  ▪ Turbocharger
		  ▪ EGR
		  ▪ Fuel system
			diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check and install a new exhaust gas temperature sensor - Post catalytic converter as required
			▪ Check DPF for mechanical integrity
			▪ Check electric throttle butterfly valve for mechanical integrity
			▪ Check engine control module for related DTCs and refer to relevant DTC index
			▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P242B-62	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 3 - Signal compare failure		▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03F8 Exhaust Gas

			NOTE: Exhaust gas temperature sensor - Post catalytic converter sensor rationality check. Temperature sensor is checked against other temperature sensors at coldstart - The engine control module detected failure when comparing two or more input parameters for plausibility - Exhaust gas temperature sensor - Post catalytic converter - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Post catalytic converter failure	Temperature Bank2 Sensor 2 - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new exhaust gas temperature sensor - Post catalytic converter as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
	P242B-64	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 3 - Signal plausibility failure		- Check engine control module for related DTCs and refer to relevant DTC index - Diagnosis of this DTC may require using the manufacturer approved diagnostic

			NOTE:
			
			Exhaust gas temperature sensor - Post catalytic converter sensor rationality check during a driving cycle is not plausible (compared against a temperature model). Exhaust temperature sensor temperature is checked during a vehicle driving cycle against a model temperature for rationality. Some engine speed/load conditions apply. If the difference between the sensor temperature and the model exceeds a defined threshold, a malfunction is raised
			
			- The engine control module detected plausibility failures
			- Other related DTCs
			  - Airpath
			  - Manifold
			- Exhaust gas temperature sensor - Post catalytic converter
			- Connector is
				system check datalogger signals
				  - 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2
				- Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
				- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
				- Check and install a new exhaust gas temperature sensor - Post catalytic converter as required
				- Using the manufacturer approved diagnostic system clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest	

		disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalytic converter failure	
P242C-16	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 3 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas temperature sensor - Post catalytic converter circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalytic converter failure	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using

			the 'Diagnosis Menu' tab and retest
P242D-17	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 3 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalytic converter circuit short circuit to power, open circuit, high resistance ▪ Charge air temperature sensor circuit open circuit, high resistance ▪ Exhaust gas temperature sensor - Post catalytic converter failure ▪ Charge air temperature sensor failure	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P244A-00	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - No sub type information	▪ Diagnostic failure check for minimum pressure differential characteristics	

			▪ Diesel particulate filter internal components are missing or destroyed	**NOTE:** If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) ▪ Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index ▪ Clear DTC and re-test ▪ Check and install a new diesel particulate filter as required	
	P244A-95	Diesel Particulate Filter Differential Pressure Too Low (Bank1) - Incorrect assembly	**NOTE:** Differential pressure sensor hoseline mounting check. Differential pressure sensor high pressure hoseline disconnected or differential pressure reading low ▪ The engine control module has detected that the component has been incorrectly installed e.g. hydraulic pipes crossed over, circuits cross wired or	**NOTE:** If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03DB Particulate Filter Differential Pressure Sensor Voltage - Bank 1	

			polarity errors ▪ In cold climates differential pressure sensor hose lines or metal pipes may be frozen ▪ Differential pressure sensor crossed hose lines ▪ Differential pressure sensor dropped top hose line ▪ Differential pressure sensor hose lines deteriorated ▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Differential pressure sensor failure	- Volts ▪ Check differential pressure sensor hose lines are not frozen ▪ Check differential pressure sensor hose lines are installed correctly ▪ Check differential pressure sensor hose lines for mechanical integrity ▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new differential pressure sensor as required ▪ Using the manufacturer approved diagnostic system carry out 'Diesel particulate filter differential pressure sensor replacement' routine (0x405B_02) ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the Diagnosis Menu tab and retest	

P244A-96	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - Component internal failure	▪ Destroyed particulate filter	**NOTE:** If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) ▪ Refer to the relevant pinpoint test in section 309-00
P2453-28	Diesel Particulate Filter Pressure Sensor A Circuit Range/Performance - Signal bias level out of range / zero adjustment failure	**NOTE:** Differential pressure sensor rationality check in afterrun. Differential pressure plausibility check during postdrive in every driving cycle. If the pressure exceeds a defined threshold, (set at 55hPa), the sensor is no longer within specification range ▪ In cold climates differential pressure sensor hose lines or metal pipes may be frozen ▪ Differential pressure sensor hoses connected incorrectly ▪ Differential pressure	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0xD922 Particulate Filter Differential Pressure - Measured ▪ Measure with engine off ▪ Check differential pressure sensor hose lines are not frozen ▪ Check differential pressure sensor hose lines are installed correctly ▪ Check differential pressure sensor hose lines for mechanical integrity ▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground,

			sensor hoses crushed, blocked, split ▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Differential pressure sensor failure	short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new differential pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
	P2453-2A	Diesel Particulate Filter Pressure Sensor A Circuit Range/Performance - Signal stuck in range	**NOTE:** Differential pressure sensor rationality check during exhaust gas flow dynamic conditions. Differential pressure sensor pressure response responds to exhaust gas flow transients ▪ Other related DTCs ▪ Contamination in the exhaust system (soot blockage) ▪ Differential pressure sensor hoses crushed, blocked, split ▪ Diesel particulate filter system mechanical integrity	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signal, 0xD922 Particulate Filter Differential Pressure - Measured ▪ Check engine control module for related DTCs and refer to relevant DTC ▪ Check exhaust system for contamination (soot blockage) ▪ Check differential pressure sensor hoses for crushed, blocked, split ▪ Check DPF system for mechanical integrity ▪ Check air intake system for	

		• Air intake system leakage • Differential pressure sensor mechanical integrity • Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance • Connector is disconnected, connector pin is backed out, connector pin corrosion • Differential pressure sensor failure	mechanical integrity • Check differential pressure sensor for mechanical integrity • Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance • Inspect connectors for signs of water ingress, and pins for damage and/or corrosion • Check and install a new differential pressure sensor as required • Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2454-16	Diesel Particulate Filter Pressure Sensor A Circuit Low - Circuit voltage below threshold	**NOTE:** Differential pressure sensor circuit continuity check. Differential pressure sensor voltage below threshold (80mV) • The engine control module measured a voltage below a specified range but not necessarily a short	• Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ◦ 0x03DB Particulate Filter Differential Pressure Sensor Voltage - Bank1 • Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short

		circuit to ground ▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Differential pressure sensor failure	circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new differential pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P2454-21	Diesel Particulate Filter Pressure Sensor A Circuit Low - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low
		▪ In cold climates differential pressure sensor hose lines or metal pipes may be frozen
		▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Differential pressure sensor failure
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0xD922 Particulate Filter Differential Pressure - Measured
			  ▪ 0x03DB Particulate Filter Differential Pressure Sensor Voltage - Bank 1 - Volts
			  ▪ Allow thaw if the vehicle was exposed to cold ambient conditions, check DPF differential pressure sensor voltage at key on is above 350mV

			(DIDx03DB) ▪ Check differential pressure sensor hose lines for mechanical integrity ▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new differential pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2455-17	Diesel Particulate Filter Pressure Sensor A Circuit High - Circuit voltage above threshold	**NOTE:** Differential pressure sensor circuit continuity check. Differential pressure sensor voltage above threshold (4990mV) ▪ The engine control module measured a voltage above a	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03DB Particulate Filter Differential Pressure Sensor Voltage - Bank1 ▪ Refer to the electrical circuit diagrams and check the differential pressure sensor

		specified range but not necessarily a short circuit to power ▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Differential pressure sensor failure	circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new differential pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	

P2455-22	Diesel Particulate Filter Pressure Sensor A Circuit High - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high
		▪ In cold climates differential pressure sensor hose lines or metal pipes may be frozen
		▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Differential pressure sensor failure
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0xD922 Particulate Filter Differential Pressure - Measured
			  ▪ 0x03DB Particulate Filter Differential Pressure Sensor Voltage - Bank 1 - Volts
			  ▪ Allow thaw if the vehicle was exposed to cold ambient conditions, check DPF differential pressure sensor voltage at key on is below	

			800mV (DIDx03DB) ▪ Check differential pressure sensor hose lines for mechanical integrity ▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new differential pressure sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2459-00	Diesel Particulate Filter Regeneration Frequency (Bank 1) - No sub type information		▪ Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03DB Particulate Filter

			NOTE: Unexpected rapid soot loading of the diesel particulate filter leading to an increased regeneration frequency. Diesel particulate filter soot mass is compared to a maximum soot mass expected during soot loading phases when the diesel particulate filter active regeneration is started. A soot mass higher that the maximum expected soot loading leads to an increased regeneration frequency - Other related DTCs - Diesel particulate filter partial blockage - Charge air system leakage - Diesel particulate filter is getting full of soot - Diesel particulate filter hoselines disconnected - Differential pressure sensor mechanical integrity - Blocked air filter - Air intake system leakage	Differential Pressure Sensor Voltage - Bank 1 - Volts - Check differential pressure sensor reading (DIDx03DB) is lower than 800mV at ignition on - 0x05CA Exhaust Turbine A Inlet Temperature - 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 - Check charge air system for leakage, split pipework, loose hose clips and damaged hoses - Check differential pressure sensor for mechanical integrity - Check air intake system for mechanical integrity - Check exhaust system upstream of diesel particulate filter for mechanical integrity - Refer to workshop manual and carry out an on road DPF regeneration drive cycle - Using the manufacturer approved diagnostic system clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest	

			▪ Exhaust system leakage upstream of diesel particulate filter	
	P2463-00	Diesel Particulate Filter Restriction - Soot Accumulation (Bank 1) - No sub type information	**NOTES:** ▪ This monitors when the Diesel Particulate Filter (DPF) is full and unable to regenerate ▪ A customer may express a concern that the amber Diesel Particulate Filter (DPF) warning indicator is illuminated on the Instrument Cluster (IC) requesting a DPF regeneration. If the regeneration is not completed the DPF warning indicator will then turn red. The DTCs **P0234-77 and P246B-00** will be stored in the Engine Control Module (ECM) memory ▪ The electric throttle air flow characteristics are incompatible with the engine calibration	▪ Check the engine control module for related DTCs and if DTCs **P0234-77 and P246B-00** are also logged GO to Pinpoint Test **A.** ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system to check datalogger signals ▪ 0x042C Diesel Particulate Filter Soot Concentration ▪ Using the manufacturer approved diagnostic system carry out a service regeneration procedure ▪ Check charge air system for leakage, split pipework, loose hose clips and damaged hoses ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest

		- Other DPF related DTCs - Diesel particulate filter maximum soot level has been reached - Charge air system leakage	
P246B-00	Vehicle Conditions Incorrect for Diesel Particulate Filter Regeneration - No sub type information	**NOTES:** Vehicle needs to be driven in Diesel Particulate Filter (DPF) regeneration conditions. Drive at approximately 70mph in 6th gear for 30min. If 70mph can not be maintained, then keep engine loaded to allow heat to be generated. Regeneration works best under high engine loads. This DTC and Amber message will only be seen if the vehicle has been driven in conditions that DO NOT allow a sufficient regeneration time period and the current soot load is too	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. The diagnostic for this DTC runs all the time. Warning lamp and message will extinguish when the soot level is at an acceptable level - Check the engine control module for related DTCs and if DTCs **P0234-77 and P246B-00** are also logged GO to Pinpoint Test **A.** - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x042C Diesel Particulate Filter Soot Concentration - Check air filter for

			high ▪ A customer may express a concern that the amber Diesel Particulate Filter (DPF) warning indicator is illuminated on the Instrument Cluster (IC) requesting a DPF regeneration. If the regeneration is not completed the DPF warning indicator will then turn red. The DTCs **P0234-77 and P246B-00** will be stored in the Engine Control Module (ECM) memory ▪ The electric throttle air flow characteristics are incompatible with the engine calibration ▪ Diesel particulate filter soot load has crossed an engineering threshold and a regeneration is required to avoid further actions being taken such as a torque limit and a red warning message. This fault will show the **amber warning "Exhaust Filter Nearly Full See**	mechanical integrity ▪ Check differential pressure sensor for hoseline correct installation ▪ Check differential pressure sensor for mechanical integrity ▪ Check air intake system for mechanical integrity ▪ Check exhaust system upstream of diesel particulate filter for mechanical integrity ▪ Refer to workshop manual and carry out an on road DPF regeneration drive cycle ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		Handbook" if during the previous ~35 minutes the vehicle has not been driven in conditions that give a successful regeneration - Diesel particulate filter high soot load. Suitable conditions must be found to allow correct regeneration of the DPF before service action is required - Continuous diesel particulate filter soot level monitoring ; if the mass exceeds upper limit; the warning is set - Diesel particulate filter is getting full of soot - Diesel particulate filter hoselines disconnected - Differential pressure sensor mechanical integrity - Blocked air filter - Air intake system leakage - Exhaust system leakage upstream of diesel particulate filter	
P246E-22	Exhaust Gas Temperature Sensor Circuit (Bank 1 Sensor 4) - Signal amplitude > maximum		- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals - 0x03F6 Exhaust Gas Temperature Bank1 Sensor 3

			NOTE: To monitor for too high temperature at the DPF outlet location. The DTC can indicate failure has occurred to the DPF during a previous regeneration. Temperature sensor temperature above threshold (874.96°C) - The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high - Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Post DPF failure - Overfilled DPF during drop to idle with active DPF regeneration on - Blocked front face of DPF	- 0x042C Diesel Particulate Filter Soot Concentration - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new exhaust gas temperature sensor - Post DPF as required - Check DPF for mechanical integrity - Check electric throttle butterfly valve for mechanical integrity - Check engine control module for related DTCs and refer to relevant DTC index - Using the manufacturer approved diagnostic system clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest

		- Stuck closed electric throttle butterfly valve during active DPF regeneration - Other related DTCs - Airpath - Turbocharger - EGR - Fuel system	
P246F-62	Exhaust Gas Temperature Sensor Circuit Range/Performance (Bank 1 Sensor 4) - Signal compare failure	**NOTE:** Exhaust gas temperature sensor - Post DPF sensor rationality check. Temperature sensor is checked against other temperature sensors at coldstart - The engine control module detected failure when comparing two or more input parameters for plausibility - Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas	- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x03F6 Exhaust Gas Temperature Bank1 Sensor 3 - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new exhaust gas temperature sensor - Post DPF as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest

			temperature sensor - Post DPF failure	

	P246F-64	Exhaust Gas Temperature Sensor Circuit Range/Performance (Bank 1 Sensor 4) - Signal plausibility failure	**NOTE:**
			Exhaust gas temperature sensor - Post DPF rationality check during a driving cycle is not plausible (compared against a temperature model). Exhaust temperature sensor temperature is checked during a vehicle driving cycle against a model temperature for rationality. Some engine speed/load conditions apply. If the difference between the sensor temperature and the model exceeds a defined threshold, a malfunction is raised
			▪ The engine control module detected plausibility failures
			▪ Other related DTCs
			  ▪ Airpath
			  ▪ Manifold
			▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to ground,
				▪ Check engine control module for related DTCs and refer to relevant DTC index
				▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
				  ▪ 0x03F6 Exhaust Gas Temperature Bank1 Sensor 3
				▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open circuit, high resistance
				▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
				▪ Check and install a new exhaust gas temperature sensor - Post DPF as required
				▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post DPF failure	
P2470-16	Exhaust Gas Temperature Sensor Circuit Low (Bank 1 Sensor 4) - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post DPF failure	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03C8 Exhaust Gas Temperature Bank1 Sensor 3 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post DPF as required ▪ Using the manufacturer approved diagnostic system clear all

			stored DTCs using the 'Diagnosis Menu' tab and retest
P2471-17	Exhaust Gas Temperature Sensor Circuit High (Bank 1 Sensor 4) - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post DPF failure	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03C8 Exhaust Gas Temperature Bank1 Sensor 3 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post DPF as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2478-84	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 1) - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check

			specified range ▪ Exhaust gas temperature sensor pre turbine circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Air induction system leakage ▪ Diesel oxidation catalyst mechanical integrity ▪ Exhaust gas temperature sensor pre turbine failure	datalogger signals ▪ 0x03BF Exhaust Gas Temperature Bank 1 Sensor 1 Voltage - Volts ▪ 0x034F Exhaust Gas Temperature Bank 1 Sensor 1 - Deg C ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage - Volts ▪ 0x05CA Exhaust Turbine A Inlet Temperature - Deg C ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor pre turbine circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check air induction system for leakage ▪ Check diesel oxidation catalyst system for mechanical integrity ▪ Check and install a new exhaust gas temperature sensor pre turbine as required ▪ Using the manufacturer approved diagnostic system clear all		

			stored DTCs using the 'Diagnosis Menu' tab and retest
P2478-85	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 1) - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Exhaust gas temperature sensor pre turbine circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Air induction system leakage - Diesel oxidation catalyst mechanical integrity - Exhaust gas temperature sensor pre turbine failure	- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x03BF Exhaust Gas Temperature Bank 1 Sensor 1 Voltage - Volts 0x034F Exhaust Gas Temperature Bank 1 Sensor 1 - Deg C 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage - Volts 0x05CA Exhaust Turbine A Inlet Temperature - Deg C - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor pre turbine circuit for short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check air induction system for leakage - Check diesel oxidation catalyst system for mechanical integrity

			- Check and install a new exhaust gas temperature sensor pre turbine as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P2479-84	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 2) - Signal below allowable range	**NOTE:**
		Rationality check at start - Exhaust gas temperature sensor temperature under threshold (-40.04°C)
		- The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
		- Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		- Connector is disconnected, connector pin is backed out, connector pin corrosion
		- Exhaust gas temperature sensor - Pre catalytic converter failure
			- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  - 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1
			  - 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage
			- Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			- Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required
			- Using the manufacturer approved diagnostic

			system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P2479-85	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 2) - Signal above allowable range	**NOTE:**
		Rationality check at start - Exhaust gas temperature sensor temperature above threshold (879.96°C)
		- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range
		- Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		- Connector is disconnected, connector pin is backed out, connector pin corrosion
		- Exhaust gas temperature sensor - Pre catalytic converter failure
			- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  - 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1
			  - 0x03E7 Exhaust Gas Temperature Bank2 Sensor 1 Voltage
			- Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			- Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required
			- Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P247A-84	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 3) - Signal below	- The engine control module has determined failures	- Diagnosis of this DTC may require using the

	allowable range	where some circuit quantity, reported via serial data, is below a specified range ▪ Exhaust gas temperature sensor - Post catalytic converter ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalytic converter failure	manufacturer approved diagnostic system check datalogger signals ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P247A-85	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 3) - Signal above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Exhaust gas temperature sensor - Post catalytic converter circuit short	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2

		circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post catalytic converter failure	▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post catalytic converter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P247B-84	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 4) - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03F6 Exhaust Gas Temperature Bank1 Sensor 3 ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open

			- corrosion - Exhaust gas temperature sensor - Post DPF failure	- circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new exhaust gas temperature sensor - Post DPF as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
	P247B-85	Exhaust Gas Temperature Out of Range (Bank 1 Sensor 4) - Signal above allowable range	- The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Exhaust gas temperature sensor - Post DPF failure	- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x03F6 Exhaust Gas Temperature Bank1 Sensor 3 - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new exhaust gas temperature sensor -

			Post DPF as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P249F-00	Excessive Time To Enter Closed Loop DPF Regeneration Control - No sub type information	**NOTE:** Diesel particulate filter temperature of the inner loop (pre turbine) did not reach target on time. Diesel particulate filter temperature of the outer loop (post catalyst) did not reach target on time ▪ Other related DTCs ▪ Exhaust gas temperature sensor - Pre turbine - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor - Pre turbine removed from gas stream ▪ Exhaust gas temperature sensor - Post catalytic converter - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor -	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CA Exhaust Turbine A Inlet Temperature ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage ▪ Check sensor for voltage plausibility ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Check sensors are installed correctly

			Post catalytic converter removed from gas stream ▪ Air induction system leakage ▪ Diesel oxidation catalyst mechanical integrity	▪ Check air induction system for leakage ▪ Check diesel oxidation catalyst for mechanical integrity ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

	P24A0-00	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low - No sub type information	**NOTE:**
			Diesel particulate filter temperature of the inner loop (pre turbine) is under temperature target. Diesel particulate filter temperature of the outer loop (post catalyst) is under temperature target
			▪ Other related DTCs
			▪ Exhaust gas temperature sensor - Pre turbine - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Exhaust gas temperature sensor - Pre turbine removed from gas stream
			▪ Exhaust gas temperature sensor - Post catalytic converter - Wiring
				▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
				  ▪ 0x05CA Exhaust Turbine A Inlet Temperature
				  ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage
				  ▪ Check sensor for voltage plausibility
				  ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2
				  ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage
				▪ Check engine control module for related DTCs and refer to relevant DTC index
				▪ Refer to the electrical circuit diagrams and check connections are

		integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor - Post catalytic converter removed from gas stream ▪ Air induction system leakage ▪ Diesel oxidation catalyst mechanical integrity	secure and wiring integrity ▪ Check sensors are installed correctly ▪ Check air induction system for leakage ▪ Check diesel oxidation catalyst for mechanical integrity ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P24A1-00	Closed Loop DPF Regeneration Control At Limit - Temperature Too High - No sub type information	**NOTE:**
		Diesel particulate filter temperature of the inner loop (pre turbine) is over temperature target. Diesel particulate filter temperature of the outer loop (post catalyst) is over temperature target
		▪ Other related DTCs
		▪ Exhaust gas temperature sensor - Pre turbine - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Exhaust gas temperature sensor - Pre turbine removed from gas stream
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0x05CA Exhaust Turbine A Inlet Temperature
			  ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage
			  ▪ Check sensor for voltage plausibility
			  ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2
			  ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage
			▪ Check engine control module for related DTCs and refer to relevant DTC index

		- Exhaust gas temperature sensor - Post catalytic converter - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor - Post catalytic converter removed from gas stream - Air induction system leakage - Diesel oxidation catalyst mechanical integrity	- Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Check sensors are installed correctly - Check air induction system for leakage - Check diesel oxidation catalyst for mechanical integrity - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P24A2-00	Particulate Filter Regeneration Incomplete Bank 1 - No sub type information	**NOTE:** Diesel particulate filter soot mass is compared to a maximum soot mass expected at the end of an active diesel particulate filter active regeneration. A soot mass higher that the maximum expected soot mass after regeneration leads to an increased regeneration frequency - Other related DTCs - Fueling	- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x03DB Particulate Filter Differential Pressure Sensor Voltage - Bank 1 - Volts - Check differential pressure sensor reading (DIDx03DB) is lower than 800mV at ignition on 0x05CA Exhaust Turbine A Inlet Temperature 0x03F8 Exhaust Gas Temperature

		- Exhaust gas temperature sensors - Differential pressure sensor - Diesel particulate filter partially blocked - Diesel particulate filter mechanical damage - Intake air induction system leakage	Bank2 Sensor 2 - Check engine control module for related DTCs and refer to relevant DTC index - Check DPF system for mechanical integrity - Check intake air induction system for leakage - Refer to the diesel particulate filter regeneration procedure and carry out a diesel particulate filter regeneration - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P24A4-97	Diesel Particulate Filter Restriction - Soot Accumulation Too High (Bank 1) - Component or system operation obstructed or blocked	- The engine control module has detected that the operation of a component is prevented by an obstruction - Other related DTCs - Air filter blocked - Contamination in the exhaust system - Differential pressure sensor hoses crushed, blocked, split - Diesel particulate filter system mechanical integrity - Blocked regeneration - Unable to reach good conditions to regenerate the DPF	- Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x042C Diesel Particulate Filter Soot Concentration - Check engine control module for related DTCs and refer to relevant DTC index - Check the condition of the air filter - Check differential pressure sensor hoses for crushed,

		▪ Customer driving routine does not allow the system to clean the particulate filter ▪ Diesel particulate filter failure	blocked, split ▪ Check DPF system for mechanical integrity ▪ Refer to the diesel particulate filter regeneration procedure and carry out a diesel particulate filter regeneration ▪ Advise customer of driving routine required to regenerate diesel particulate filter as stated in the vehicle handbook ▪ Check and install a new diesel particulate filter as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P24C2-62	Exhaust Gas Temperature Measurement System - Multiple Sensor Correlation Bank 1 - Signal compare failure		▪ Check engine control module for related DTCs and refer to relevant DTC index
			▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
			  ▪ 0x05CA Exhaust Turbine A Inlet Temperature
			  ▪ 0x03F7 Exhaust Gas Temperature Bank2 Sensor 1

			NOTE: Exhaust temperature sensor - Cross temperature sensor check failed at coldstart. More than one temperature sensor is not within a plausible range when comparing each other. Multiple temperature sensor temperature at least 50°C above other exhaust temperature sensors temperature after 90min engine off time - The engine control module detected failure when comparing two or more input parameters for plausibility - Other related DTCs - Exhaust gas temperature sensor - Turbine temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor - Pre catalytic converter circuit short circuit to ground, short circuit to power, open	- 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 - 0x03F6 Exhaust Gas Temperature Bank1 Sensor 3 - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Turbine temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Pre catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post catalytic converter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to	

		circuit, high resistance ▪ Exhaust gas temperature sensor - Post catalytic converter circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Turbine temperature sensor failure ▪ Exhaust gas temperature sensor - Pre catalytic converter failure ▪ Exhaust gas temperature sensor - Post catalytic failure ▪ Exhaust gas temperature sensor - Post DPF failure	ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Turbine temperature sensor as required ▪ Check and install a new exhaust gas temperature sensor - Pre catalytic converter as required ▪ Check and install a new exhaust gas temperature sensor - Post catalytic converter as required ▪ Check and install a new exhaust gas temperature sensor - Post DPF as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

Pre Catalyst Oxygen Sensor Adaption Process

Pre catalyst oxygen sensor adaption cycle is NOT possible while DTCs are logged. Using the manufacturer approved diagnostic system, carry out FIT NEW PARTICULATE FILTER PROCESS only and clear the DTCs. After completing FIT NEW PARTICULATE FILTER PROCESS continue with pre catalyst oxygen sensor adaption cycle as follows:-1. Idle vehicle for 10 minutes 2. Set car in command shift 3rd gear 3. Accelerate to 3800RPM (where achievable) and overrun / coast down without braking until revs drop below 1500 RPM 4. Repeat step 3 a further 3 times 5. Check for any DTCs. If adaption has failed a DTC will be evident and the sensor will require replacement 6. Check oxygen sensor adaption is now complete. Using the manufacturer approved diagnostic system clear the DTC and re-test. Return vehicle to the customer

Differential Pressure Sensor Adaption Process

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

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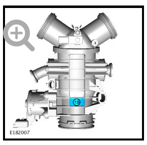
DIESEL PARTICULATE FILTER WARNING DISPLAYED ON THE INSTRUMENT CLUSTER, WITH DTCS P246B-00 - P2463-00 - P0234-77 STORED IN THE ECM

NOTE:

A new electric throttle should only be installed when it is marked with a date stamp of 13 or 14 (this denotes that the part was manufactured in 2013/2014). If the air intake shut off throttle has a date stamp of 15 then it must not be replaced

CAUTION:

This procedure requires SDD DVD142.05 and Calibration File 214 installed or later

PINPOINT TEST A : DIESEL PARTICULATE FILTER WARNING DISPLAYED ON THE INSTRUMENT CLUSTER, WITH DTCS P246B-00 - P2463-00 - P0234-77 STORED IN THE ECM	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: CHECK FOR DTCS (OTHER THAN P246B-00/P2463-00/P0234-77)	
	1 Using the manufacturer approved diagnostic system check for charge air system related DTCs stored in the engine control module
	Are any charge air system related DTCs stored in the engine control module? **Yes** The charge air system related DTCs must be rectified before proceeding to the next step. Refer to the relevant DTC index and carry out repair actions. **No** Proceed to next step
A2: CHECK DATE STAMP OF THE ELECTRIC THROTTLE	
	1 Remove the engine cover, (refer to TOPIx, Workshop Manual, section 501-05).
	2 Inspect the date stamp on the electric throttle

	Is the date stamp on the electric throttle 13 or 14? **Yes** Replace the electric throttle, (refer to TOPIx Workshop Manual, section 304-04). Install the engine cover, (refer to TOPIx, Workshop Manual, section 501-05). Using the manufacturer approved diagnostic system carry out the 'Powertrain set up - Throttle valve actuator' routine Proceed to next step **No** Proceed to next step

A3: CARRY OUT DIAGNOSTIC PROCEDURE

	1 Connect the approved battery support unit.
	2 Using the manufacturer approved diagnostic system connect to the vehicle and begin a new diagnostic session, by reading the vehicle identification number (VIN) for the current vehicle and initiating the data collect sequence.
	3 Follow the manufacturer approved diagnostic system prompts.
	4 Diagnosis of these DTCs will require using the manufacturer approved diagnostic system check datalogger signal – (0x042C) 'Diesel Particulate Filter Soot Concentration - Inferred Using Differential Pressure' measured in g/l –.
	5 Using manufacturer approved diagnostic system data logger record the Diesel Particulate Filter Soot Concentration value.
	Is the Diesel Particulate Filter Soot Concentration value greater than 6 grams **Yes** Proceed to next step **No** Disconnect SDD Disconnect battery support unit Release vehicle to customer

A4: CARRY OUT DIAGNOSTIC PROCEDURE - DIESEL PARTICULATE FILTER DYNAMIC REGENERATION

	1 Using the manufacturer approved diagnostic system carry out the 'Powertrain - Diesel particulate filter dynamic regeneration' routine
	2 Using manufacturer approved diagnostic system data logger record the Diesel Particulate Filter Soot Concentration value.
	Is the Diesel Particulate Filter Soot Concentration value greater than 6 grams **Yes** Repeat the Powertrain - Diesel particulate filter dynamic regeneration' routine until the Diesel Particulate Filter Soot Concentration value is less than 6 grams

	No Disconnect SDD Disconnect battery support unit Release vehicle to customer

DIESEL PARTICULATE FILTER SOOT LEVEL INFORMATION

SOOT LEVEL SYMPTOM	VALUES
Vehicle is still able to perform a drive cycle diesel particulate filter regeneration	23 grams soot
Amber warning indicator is displayed - Customer can still perform a drive cycle diesel particulate filter regeneration	27 grams soot
Red warning indicator is displayed - Visit the Dealer	30.5 grams soot
Diesel particulate filter regeneration is NOT possible - Replace diesel particulate filter	31.5 grams soot

DIESEL PARTICULATE FILTER REGENERATION PROCEDURE

If DPF FULL SEE HANDBOOK appears in the message center, carry out the following procedure

CAUTIONS:

- The regeneration procedure produces high temperatures in the diesel particulate filter. Heat can be felt radiating from beneath the vehicle, which is normal and not a cause for concern. However, the vehicle should not be parked over combustible material, particularly during dry weather. The heat generated could be sufficient to start a fire when in close proximity to combustible material such as long dry grass, paper etc
- At all times during this procedure you should observe all relevant speed limits, laws, and regulations

NOTES:

- The ideal speed and conditions for regeneration are 100 km/h (62 mph) ~ 120 km/h (75 mph), in Drive. Keeping a constant speed enables the diesel particulate filter to regenerate more efficiently. It is therefore recommended that cruise control is used to achieve this, if possible
- When driving off-road during the regeneration process, greater accelerator pedal use may be required

1. Drive the vehicle until the engine reaches normal operating temperature. The engine should **NOT** be left idling to achieve working temperature

2. Drive the vehicle for a further twenty minutes, keeping the vehicle at a constant speed between 75 km/h (45 mph) and 120 km/h (75 mph)

3. If regeneration is successful the warning message will be extinguished, once the message is extinguished please keep driving for 10 minutes to ensure that the diesel particulate filter is completely clean

4. If the message remains repeat the process

DIAGNOSTIC PROCEDURES TO BE USED IF DTC P244A-00, P244A-95 OR P244A-96 IS LOGGED IN THE ENGINE CONTROL MODULE

PINPOINT TEST B : P244A-00, P244A-95 OR P244A-96	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: PRE-CHECK (ENGINE CONTROL MODULE SOFTWARE)	
	1 Check the latest relevant level of software is installed to the engine control module
	Is the latest relevant level of software installed to the engine control module? **Yes** Proceed to the next step **No** 1. Using the manufacturer approved diagnostic system, re-configure the engine control module with the latest level software2. Proceed to the next step
B2: PRE-CHECK (LOGGED DTCS)	
	1 Check that there are not other engine control module DTCs logged
	Are other engine control module DTCs logged? **Yes** 1. Refer to the relevant DTC index and rectify as required2. Clear the DTC and retest. When using the manufacturer approved diagnostic system clear all stored DTCs from the diagnosis menu tab3. Proceed to the next step **No** Proceed to the next step
B3: PRE CHECK (DIFFERENTIAL PRESSURE SENSOR ADAPTION)	
	1 Check the differential pressure sensor offset value. The following conditions must be achieved to allow the differential pressure sensor offset value to be learnt - stored: Using the manufacturer approved diagnostic system, monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park - set ignition status to off. Wait 30 seconds for the engine control module to power down, learn - store differential pressure sensor offset value. This process must be carried

	out six times, to allow a large negative offset value to adapt back to 0 hPa
	Has the differential pressure sensor adaption offset value check been carried out? **Yes** Proceed to the next step **No** 1. Carry out the differential pressure sensor adaption offset value check2. Proceed to the next step

B4: DIFFERENTIAL PRESSURE SENSOR, HOSE CHECKS

	1 Check the differential pressure sensor hoses for splits, crushing, connectivity, blockages or contamination
	Were the differential pressure sensor hoses split, crushed, blocked or contaminated? **Yes** 1. Rectify as required2. Clear the DTC and retest. When using the manufacturer approved diagnostic system clear all stored DTCs from the diagnosis menu tab **No** Proceed to the next step

B5: DIFFERENTIAL PRESSURE SENSOR, CONNECTOR CHECKS

	1 Check the differential pressure sensor connector for disconnected, connector pin is backed out, connector pin corrosion
	Was the differential pressure sensor connector disconnected, connector pins backed out, connector pin corroded **Yes** 1. Rectify as required 2. Clear the DTC and retest. When using the manufacturer approved diagnostic system clear all stored DTCs from the diagnosis menu tab **No** Proceed to the next step

B6: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT

	NOTE: The following checks are designed to verify the differential pressure sensor, pressure/voltage transfer function and will determine if the differential pressure sensor operates within tolerance. The checks will also check for leakage within the differential pressure sensor assembly and pipework. The measured voltages of the differential pressure sensor will be the result of pressure being applied to the reference input and high pressure input on the differential pressure sensor.

	1 Connect a vehicle battery stabilizer
	2 Set the vehicle to park
	3 Set ignition to 'ON' engine 'OFF'
	4 Allow voltage to stabilize
	5 Disconnect both the rubber hoses from the metal pipes of the diesel particulate filter (the rubber hoses remain connected to the differential pressure sensor).
	6 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 0.49 volts & 0.51 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as required

B7: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED ZERO)

	1 Install the manufacturer approved hand vacuum pump to the reference pipe and differential pressure sensor. Ensure the gauge is reading zero value for vacuum and pressure
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 0.485 volts & 0.515 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as required

B8: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 5 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 5 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 1.35 volts & 1.47 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described

	in test step B3

B9: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 10 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 10 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 2.30 volts & 2.45 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B10: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 15 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 15 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 3.30 volts & 3.45 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B11: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 20 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 20 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 4.25 volts & 4.40 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B12: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 25 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 25 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 4.75 volts & 4.90 volts **Yes** Release the vacuum and remove the hand vacuum pump from the reference pipe and differential pressure sensorProceed to the next step **No** Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B13: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED ZERO)

	1 Install the manufacturer approved hand vacuum pump to the high pressure pipe and differential pressure sensor. Ensure the gauge is reading zero value for vacuum and pressure
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 0.490 volts & 0.515 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as required

B14: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 20KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 20 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 1.60 volts & 1.75 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as required

B15: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 40KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 40 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 2.70 volts & 2.85 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as required

B16: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 60KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 60 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 3.80 volts & 3.90 volts **Yes** Proceed to the next step **No** Check and install a new differential pressure sensor as required

B17: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 80KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 80 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – **Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -**
	Does the voltage measure between 4.0 volts & 4.89 volts **Yes** The differential pressure sensor is OK and should not be replaced Carry out the differential pressure sensor adaption offset value learn as described in test step B3 **No** Check and install a new differential pressure sensor as required Carry out the differential pressure sensor adaption offset value learn as described in test step B3

DTC INDEX

For a list of DTCs that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 2, DTC: P2380-23 to U300F-00 (100-00 General Information, Description and Operation).